Bridging the Gap in LAI-PrEP Implementation

Reconceptualizing the HIV prevention cascade and computationally validating clinical decision support at the UNAIDS 21.2 Million target scale.

Published in Viruses (MDPI) • February 13, 2026
DOI: 10.3390/v18020237

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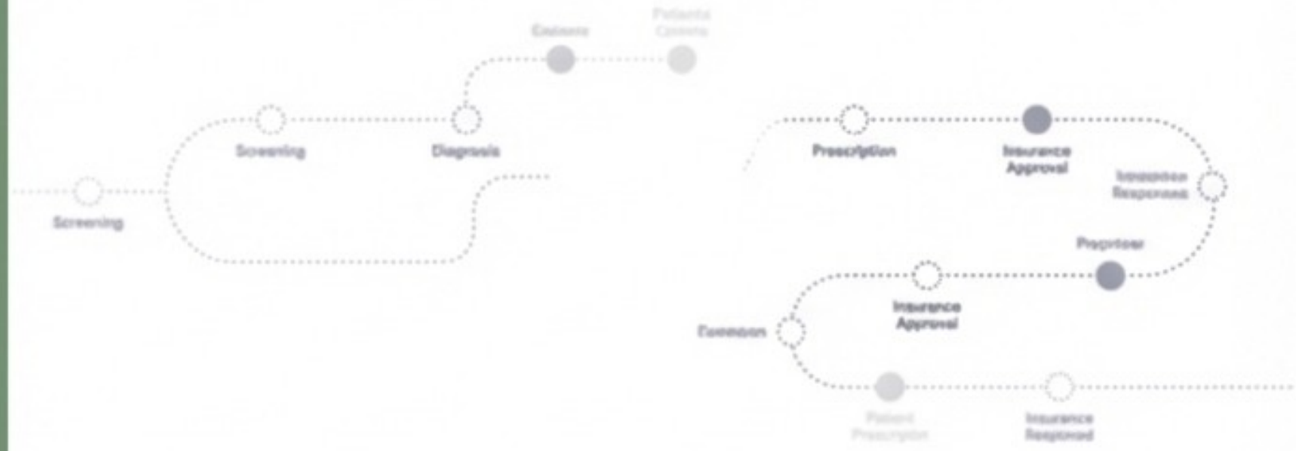
A Clinical Miracle Facing a Real-World Paradox

>96% Efficacy

The Promise: Superior efficacy validated across >15,000 participants in landmark clinical trials (HPTN 083, HPTN 084, PURPOSE-1, PURPOSE-2).



The Paradox: Despite extraordinary clinical efficacy, real-world implementation faces a critical bottleneck before the drug is ever administered.

The 47.1% Pre-Initiation Attrition Blind Spot

The Bridge Period: A mandatory 2-8 week temporal and procedural gap between a clinical prescription and the first injection, required to confirm HIV-negative status.

The Attrition Cliff: In real-world implementation, 47.1% of prescribed individuals never receive their first injection.

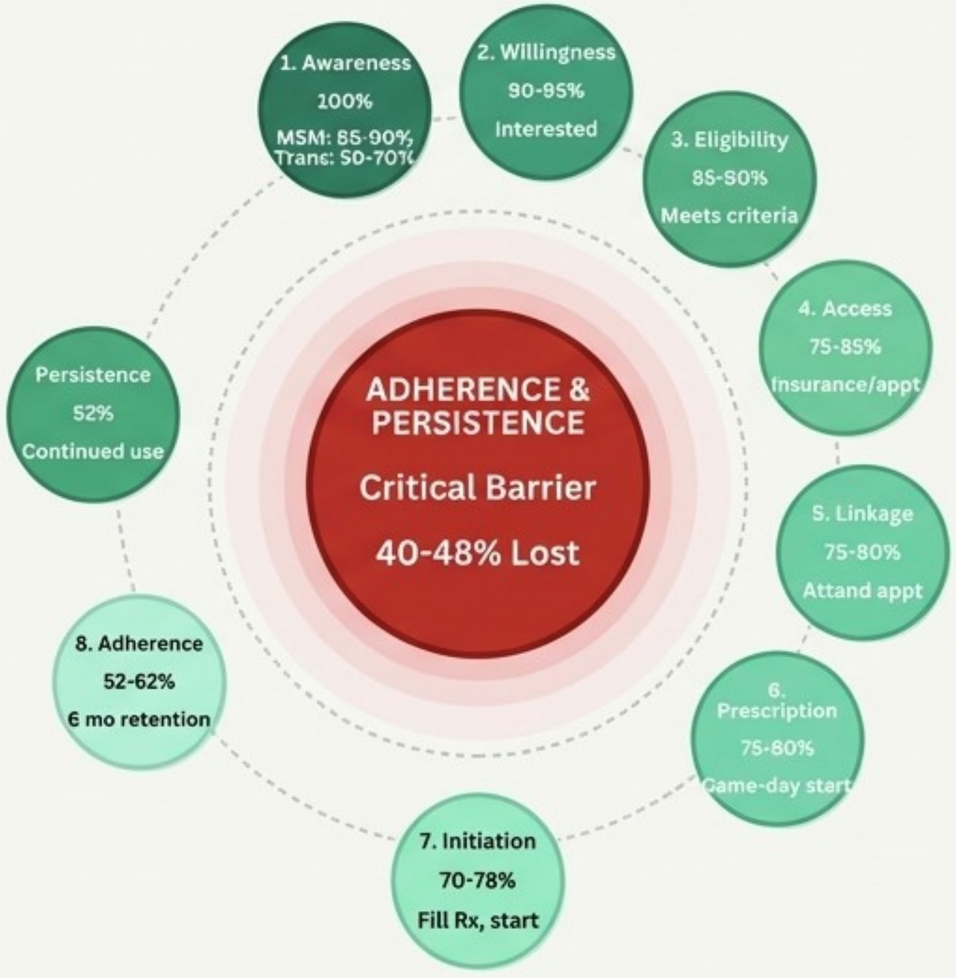
The Vulnerability: Patients remain highly motivated but entirely unprotected while navigating HIV testing delays, insurance authorizations, and appointment coordination.

Shifting the Paradigm from Adherence to Infrastructure

Oral PrEP: Same-day start. Primary failure is post-initiation adherence (40-48% lost).

LAI-PrEP: Mandatory diagnostic delay. Primary failure is pre-initiation bridge navigation (47.1% lost).

The Persistence Payoff: Successful bridge navigation yields an 81-83% persistence rate.



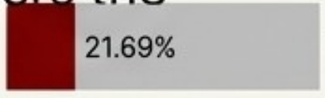
Structural Barriers Drive a Profound Equity Crisis

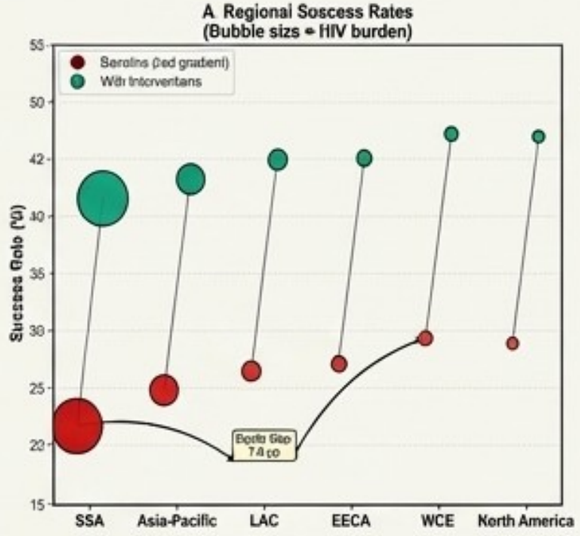
Bridge period attrition relies heavily on structural resources, severely impacting marginalized groups who lack them:

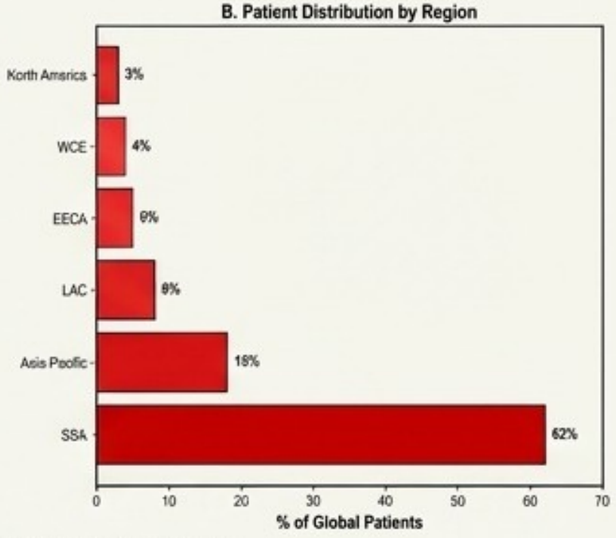
People Who Inject Drugs (PWID): Only a 10.36% baseline success rate due to housing instability and criminalization fears.

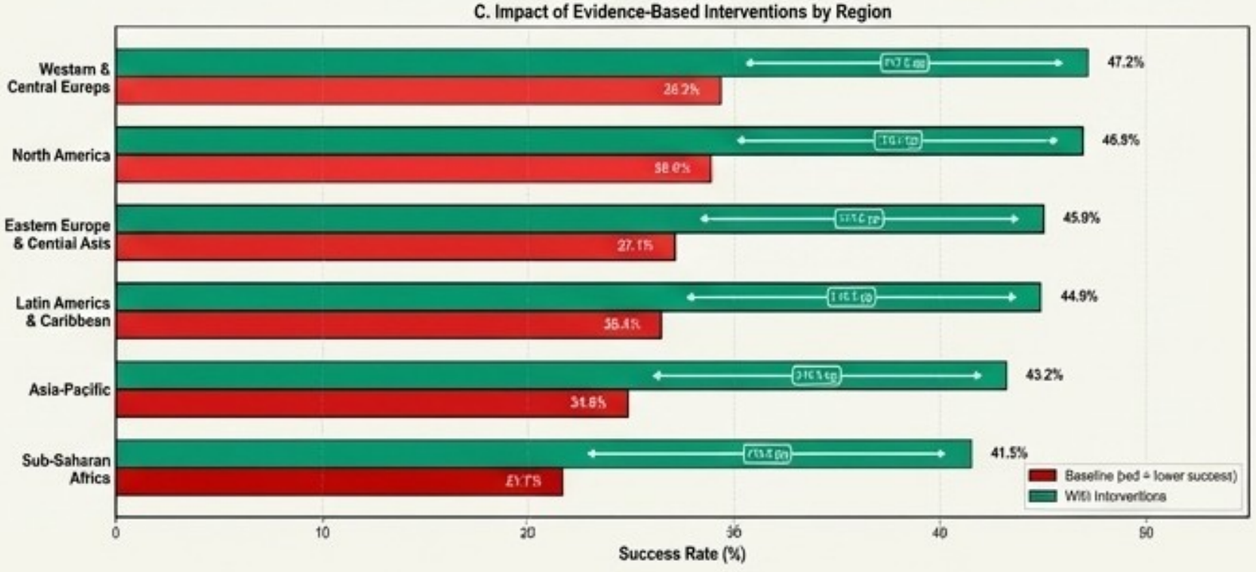
Adolescents (16-24): Only a 16.34% baseline success rate due to privacy concerns and comple healthcare navigation.

Sub-Saharan Africa (SSA): Serves 62% of global PrEP patients, yet suffers the lowest regional baseline success rate (21.69%).









Synthesizing Evidence into Clinical Decision Support

An Engine of Twenty-One Evidence-Based Interventions

The algorithm deploys strategies across four primary mechanisms:

Eliminate the Bridge

Same-day oral-to-injectable switching protocols (+35% absolute improvement).

Compress the Bridge

Rapid 24-48h laboratory turnarounds and point-of-care HIV testing.

Navigate the Bridge

Dedicated patient and peer navigation programs for complex logistics.

Remove Barriers

Telemedicine integration and bundled episode-based payment models.

Computational Validation at the UNAIDS Global Target Scale

21.2 MILLION

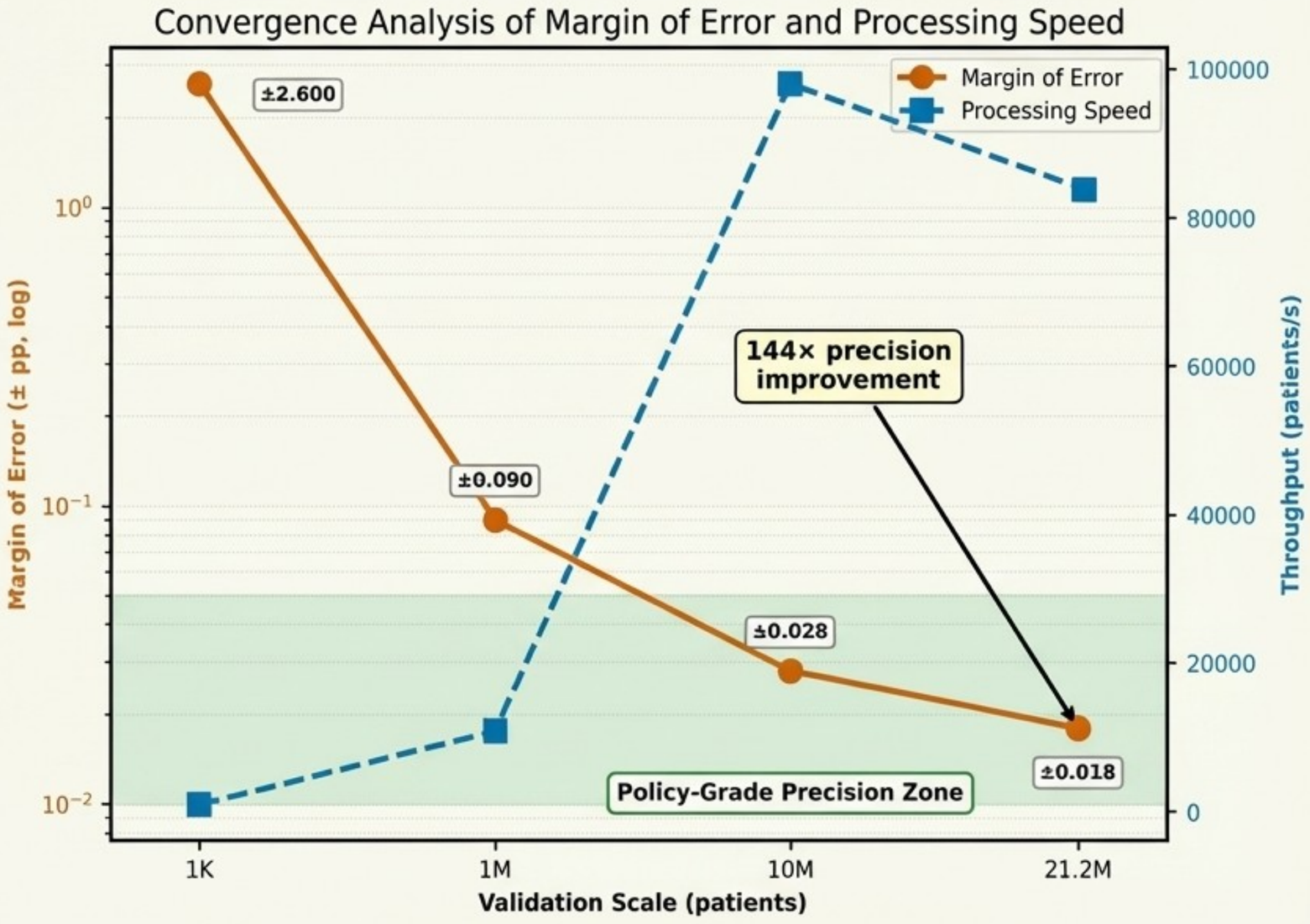
Unprecedented Scale:
Progressively validated from 1,000 up to exactly 21.2 Million simulated patients—matching the UNAIDS 2025 target.

Stress-Tested:
Comprehensive unit testing across 18 clinical edge cases (e.g., maximum barriers, extremes of age).

Reliability:
Achieved a 100% test pass rate, proving mathematical soundness across the full clinical spectrum.

Achieving Policy-Grade Statistical Precision

- As the simulation scaled, precision improved by 144x.
- **Ultimate Precision:** Reached a margin of error of ±0.018 percentage points at the 21.2M scale.
- **The Implication:** This is not just a clinical estimate; it is policy-grade statistical precision suitable for informing WHO and UNAIDS international guidelines and resource allocation.



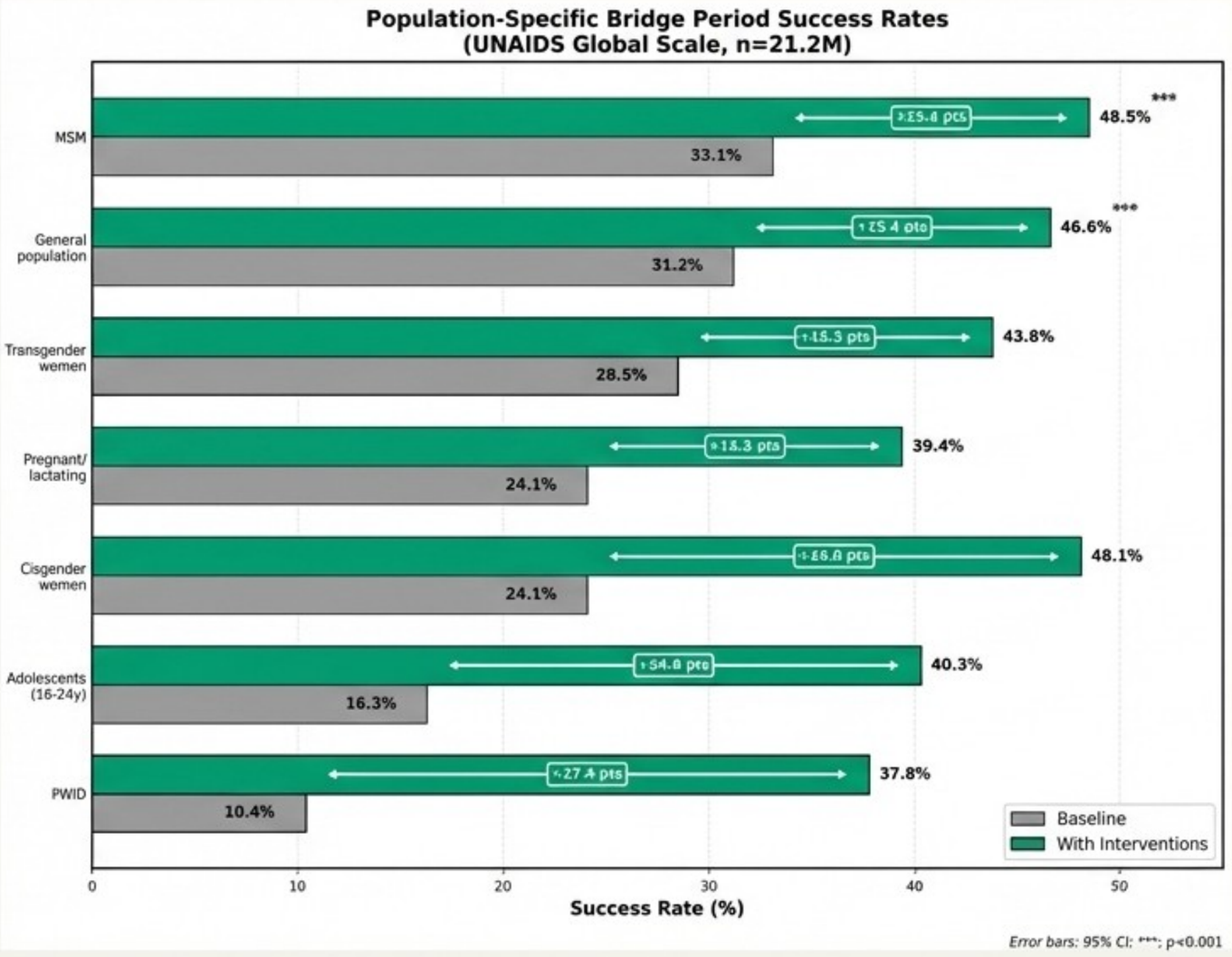
Narrowing the Health Equity Gap for Vulnerable Populations

Evidence-based interventions disproportionately benefit the populations facing the highest structural barriers:

People Who Inject Drugs (PWID): Experienced a +265% relative improvement in successful transitions.

Adolescents: Experienced a +147% relative improvement.

Systemic Impact: Systematic implementation support narrows health equity gaps at a global scale.



Translating Algorithmic Gains into Global Health Impact

At the UNAIDS 21.2M target scale, deploying this Clinical Decision Support tool elevates the global success rate from **23.96%** to **43.50%**.

4.1 Million Additional Successful Transitions

19.5% of total (23.4% of UNAIDS gap)

100,000 HIV Infections Prevented Annually

(500,000+ over 5 years)

The Economic Case for Bridge Period Management

11:1
Return on Investment

$40 Billion Lifetime Treatment Costs Saved

The prevention of 100,000 annual infections corresponds to an estimated **$40 Billion** in averted lifetime HIV treatment costs.

Generates an 11:1 cumulative **return on investment** for health systems and payers over a five-year period.

Fixing Healthcare Systems Instead of Blaming Patients

LAI-PrEP failure is a structural vulnerability, not an individual behavioral failure.

Conservative initiation protocols serve a vital safety function, but we must streamline the infrastructure to retain motivated individuals during the mandatory wait.

We must shift our focus from patient non-adherence to healthcare delivery optimization.

Partner with Nyx Dynamics for Pilot Implementation

The computational validation is complete. The next step is translating this potential into clinical reality through rigorous prospective validation. Nyx Dynamics is actively seeking pilot site placement to integrate this Clinical Decision Support tool into real-world clinical workflows. Interested major academic medical centers, clinicians, and researchers:

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Demidont AC. Viruses 2026, 18, 237. DOI: 10.3390/v18020237